

Systematic Trading Strategies with Machine Learning Algorithms

Introduction to Unsupervised Learning Techniques



April 22, 2026

Feature Importance Analysis

- Decision Trees

- In-sample Feature Importance Analysis

- Out-of-Sample Feature Importance Analysis

Introducing Unsupervised Learning Algorithms

- Clustering methods using the K-means algorithm

- Introducing Gaussian Mixture Models

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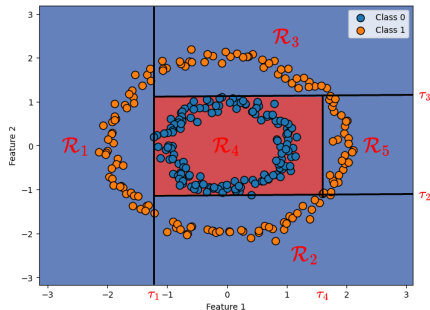
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Decision trees create recursive binary partitioning of the feature space.

- ▶ Decision tree algorithm:
 - ▶ Divides the input space into regions
 - ▶ Each split based on a single feature
 - ▶ Predictions are made based on region assignment
- ▶ *The example shows classification on ring-shaped data using simple thresholds.*



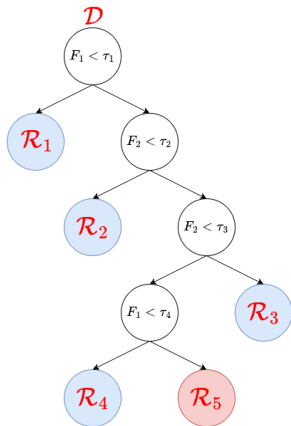
How decision trees map inputs to outputs:

- ▶ A decision tree maps input $F \in \mathbb{R}^d$ to output y using binary decision rules:
 - ▶ Each node has a splitting rule
 - ▶ Each leaf node is associated with an output value
- ▶ Each splitting rule is of the form:

$$h(F) = \mathbf{1}\{F_j > \tau\}$$

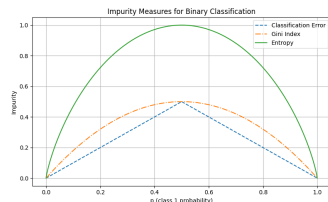
for some dimension j of F and $\tau \in \mathbb{R}$

- ▶ Using these transition rules, a path to a leaf node gives the prediction
- ▶ The leaves define regions called $\mathcal{R}_m$



Impurity measure

- ▶ For all $F \in \mathcal{R}$, let p_k be empirical fraction labeled k in this region.
- ▶ Measures of impurity for region $\mathcal{R}$:
 1. Classification error: $1 - \max_k p_k$
 2. Gini index: $\sum_{k=1}^K p_k(1 - p_k)$
 3. Entropy: $-\sum_{k=1}^K p_k \log p_k$
- ▶ *Impurity is maximized when classes are evenly distributed.*
- ▶ *Impurity is minimized when a region contains only one class.*

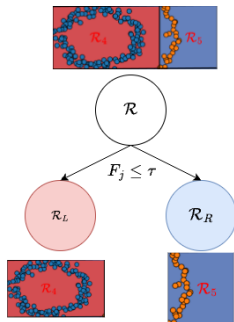


- **Information Gain** for feature j and threshold τ :

$$IG(j, \tau) = I(\mathcal{R}) - \frac{|\mathcal{R}_L|}{|\mathcal{R}|} I(\mathcal{R}_L) - \frac{|\mathcal{R}_R|}{|\mathcal{R}|} I(\mathcal{R}_R)$$

Where:

- $\mathcal{R}_L = \{F \in \mathcal{R} : F_j \leq \tau\}$
 - $\mathcal{R}_R = \{F \in \mathcal{R} : F_j > \tau\}$
-
- The DT algorithm:
 1. For each feature j and possible threshold τ , compute $IG(j, \tau)$
 2. Select feature j^* and threshold τ^* that maximize IG
 3. Split node and create child regions $\mathcal{R}_L$ and $\mathcal{R}_R$
 4. Recursively apply to each child node until stopping criteria met



Algorithm Decision Tree Learning Algorithm

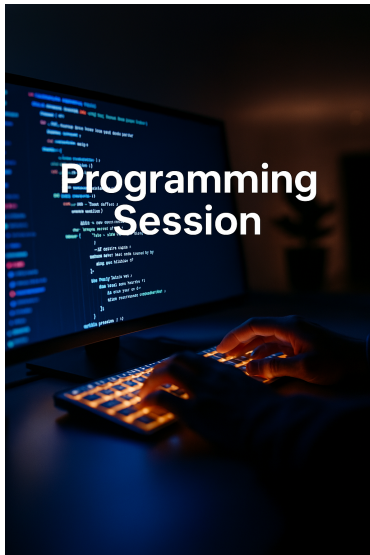
Require: Training data $\{(F_i, y_i)\}_{i=1}^n$, stopping criteria

Ensure: Decision tree T

- 1: Initialize tree with single root node containing all data
 - 2: **while** nodes can be split and stopping criteria not met **do**
 - 3: **for** each leaf node with region $\mathcal{R}$ **do**
 - 4: Find (j^*, τ^*) that maximizes:
 - 5:
$$IG(j, \tau) = I(\mathcal{R}) - \frac{|\mathcal{R}_L|}{|\mathcal{R}|} I(\mathcal{R}_L) - \frac{|\mathcal{R}_R|}{|\mathcal{R}|} I(\mathcal{R}_R)$$
 - 6: Where $\mathcal{R}_L = \{F \in \mathcal{R} : F_j \leq \tau\}$ and $\mathcal{R}_R = \{F \in \mathcal{R} : F_j > \tau\}$
 - 7: Split node using rule $F_{j^*} > \tau^*$
 - 8: **end for**
 - 9: **end while**
 - 10: Assign prediction to each leaf node (majority class)
 - 11: **return** T
-

Quiz Time!

Click here to take the quiz



Optional Programming Session: Implementation of the DT algorithm

- ▶ *Click here to access the programming session*

Content:

- ▶ Build a Decision Tree classifier from scratch.
- ▶ Recursively split data; leaf nodes store class predictions.
- ▶ Generate non-linear data and compare Logistic Regression vs. Decision Tree decision boundaries.

Feature Importance Analysis

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In-sample Feature Importance Analysis

Out-of-Sample Feature Importance Analysis

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Feature Importance Analysis in Random Forests

- ▶ Introduced by *Breiman [2001]* for measuring feature importance in tree-based models.
- ▶ Based on the accumulated decrease in impurity across all trees.

Pros

- ▶ Fast computation (calculated during training)
- ▶ Default method in most libraries
- ▶ Provides interpretable values (0-1)

Cons

- ▶ In-sample measure only.
- ▶ Can assign importance to noise.
- ▶ Biased toward high cardinality features

Algorithm Mean Decrease Impurity (MDI) for Random Forests

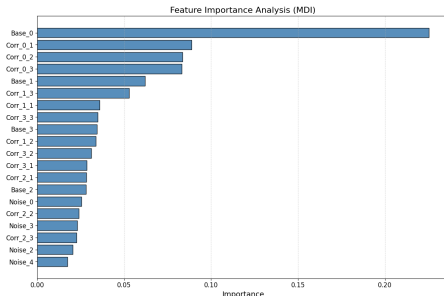
Require: Trained random forest model with T trees

Ensure: Feature importance scores $\{\text{MDI}_j\}_{j=1}^d$

- 1: Initialize importance scores: $\text{IMP}_j = 0$ for all features j
 - 2: **for** each tree $t = 1$ to T in the forest **do**
 - 3: **for** each internal node n in tree t **do**
 - 4: Identify the feature F_j used for splitting at node n
 - 5: Let w_n be the proportion of samples reaching node n
 - 6: Calculate information gain $IG_n(j, \tau_n)$
 - 7: Update importance: $\text{IMP}_j = \text{IMP}_j + w_n \cdot IG_n(j, \tau_n)$
 - 8: **end for**
 - 9: **end for**
 - 10: Compute $\text{MDI}_j = \frac{\text{IMP}_j}{\sum_{k=1}^d \text{IMP}_k}$ for all j
 - 11: **return** $\{\text{MDI}_j\}_{j=1}^d$
-

Mean Decrease Impurity

- ▶ Generated dataset with 1000 samples
- ▶ 4 feature clusters with 4 features each:
 - ▶ One base feature per cluster (Base_i)
 - ▶ Three correlated features per cluster (Corr_{i-j})
- ▶ 5 pure noise features (Noise_i)
- ▶ Ground truth: Only "Base" features directly influence the target



- ▶ **MDI Interpretation:**
- ▶ Importance is split among correlated features in each cluster
- ▶ Some noise features receive non-zero importance

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Out-of-Sample Feature Importance

- ▶ Measures the decrease in model performance when a feature is randomly shuffled.
- ▶ Intuition: If shuffling a feature decreases performance significantly, that feature is important.

Pros

- ▶ **Out-of-sample:** Evaluates on validation data.
- ▶ **Model-agnostic:** Works with any machine learning model.
- ▶ **Metric-flexible:** Compatible with any performance metric.

Cons

- ▶ Computationally expensive.
- ▶ Results vary with different random permutations.
- ▶ Sensitive to feature correlation.

Algorithm Permutation Feature Importance (PFI)

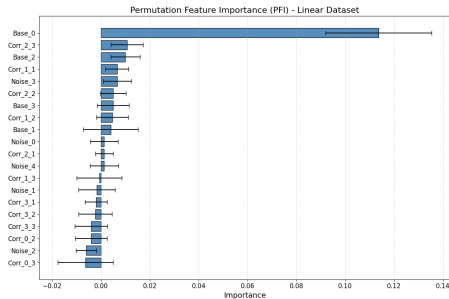
Require: Fitted model m , validation data D , repetitions K

Ensure: Feature importance scores $\{PFI_j\}_{j=1}^d$ with stds $\{\sigma_j\}_{j=1}^d$

- 1: Compute reference score s of model m on data D
- 2: **for** each feature F_j (column of D) **do**
- 3: Initialize array $scores_j$ of length K
- 4: **for** each repetition k in $1, \dots, K$ **do**
- 5: Randomly shuffle column j of dataset D to generate corrupted version $\tilde{D}_{k,j}$
- 6: Compute score $s_{k,j}$ of model m on corrupted data $\tilde{D}_{k,j}$
- 7: Store in array: $scores_j[k] = s - s_{k,j}$
- 8: **end for**
- 9: Compute mean importance PFI_j and standard deviation σ_j from array $scores_j$
- 10: **end for**
- 11: **return** $\{PFI_j\}_{j=1}^d$ and $\{\sigma_j\}_{j=1}^d$

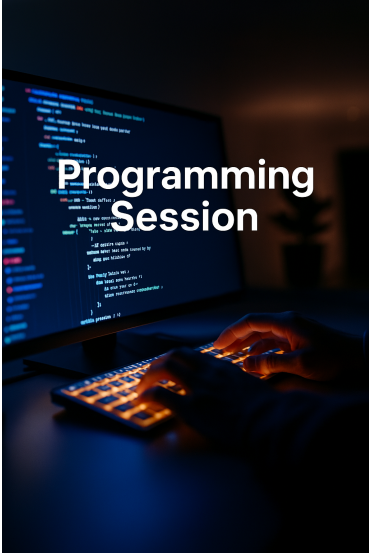
Permutation Feature Importance Analysis

- ▶ Generated dataset with 1000 samples
- ▶ 4 feature clusters with 4 features each:
 - ▶ One base feature per cluster (Base_i)
 - ▶ Three correlated features per cluster (Corr_{i-j})
- ▶ 5 pure noise features (Noise_i)
- ▶ Ground truth: Only "Base" features directly influence the target



- ▶ **PFI Interpretation:**
- ▶ More robust to noise features than MDI.
- ▶ Importance is underestimated for all correlated features

- ▶ **Issue:** When features are correlated, permuting one doesn't fully remove its information (it's still available through the other).
- ▶ This results in smaller performance drops, leading to underestimated importance scores.
- ▶ Consequence: correlated features may appear unimportant, even when jointly they matter.
- ▶ **Relevance to finance:** Feature are often highly correlated, making this a critical issue in systematic trading strategies.
- ▶ **Mitigation strategies:**
 - ▶ **Clustering:** group correlated features, select representatives, and perform feature importance at the cluster level.
 - ▶ **Dimensionality reduction:** use PCA or autoencoders to extract uncorrelated components before computing importance.



Programming Session

Programming Session 2: Sections 1 and 2

- ▶ Section 1: Generating Synthetic Data.
- ▶ Section 2: Feature Importance Analysis on Correlated Features.
- ▶ *Click here to access the programming session*

Solution will be posted tonight on the GitHub page.

- ▶ *Click here to access the GitHub Page*

Feedback Poll

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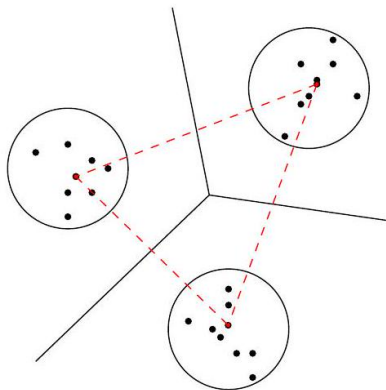
Given a data set

$X = (x_1, \dots, x_n) \in \mathbb{R}^{n \times p}$ where:

- ▶ n is the number of observations
- ▶ p is the number of features

Goal: Separate data into K clusters by learning:

- ▶ Centroids of each cluster
 $\{c_1, \dots, c_K\} \in \mathbb{R}^{p \times K}$
- ▶ Assignment function
 $\Psi : \{x_1, \dots, x_n\} \rightarrow \{1, \dots, K\}$
- ▶ Meaning: sample x_i belongs to class $\Psi(x_i)$



A simple representation of clustering
($n = 25, p = 2, K = 3$)

Algorithm The K-means Algorithm

Require: A data set $X = \{x_1, \dots, x_n\}$ ($x_i \in \mathbb{R}^p$)

Ensure: An assignment function Ψ^* and the associated centroids $c_1^*, \dots, c_K^*$.

- 1: Initialization: Choose $c_1, \dots, c_K$ in X at random
 - 2: **repeat**
 - 3: **for** $i = 1 \dots n$ **do**
 - 4: $\Psi(x_i) \leftarrow \arg \min_{k \in \{1, \dots, K\}} \|x_i - c_k\|^2$
 - 5: **end for**
 - 6: **for** $k = 1 \dots K$ **do**
 - 7:
$$c_k \leftarrow \frac{1}{\sum_{i=1}^n \mathbb{1}(\Psi(x_i) = k)} \sum_{i=1}^n \mathbb{1}(\Psi(x_i) = k) x_i$$
 - 8: **end for**
 - 9: **until** convergence
 - 10: **return** $\Psi^*, c_1^*, \dots, c_K^*$
-

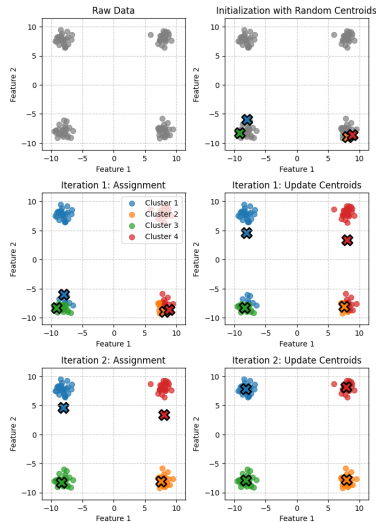
Distortion Measure:

Let Ψ be an assignment function and $c = (c_1, \dots, c_K)$ be centroids.

$$J(\Psi, c) = \frac{1}{n} \sum_{i=1}^n \|x_i - c_{\Psi(x_i)}\|^2$$

Key Properties:

- ▶ K-means monotonically decreases the distortion
- ▶ This guarantees convergence to a local minimum
- ▶ The algorithm stops after a finite number of steps
- ▶ *Optional: Click here for the proof*



K-means visualization

- ▶ Determining the appropriate number of clusters (K) is a critical challenge in cluster analysis
- ▶ Several validation metrics help identify the optimal K :
 - ▶ **Silhouette Score** - Measures cohesion and separation
 - ▶ **Calinski-Harabasz Index** - Ratio of between-cluster to within-cluster variance
 - ▶ **Davies-Bouldin Index** - Ratio of within-cluster scatter to between-cluster separation
- ▶ Approach: Run K-means with different K values, evaluate metrics, select optimal K

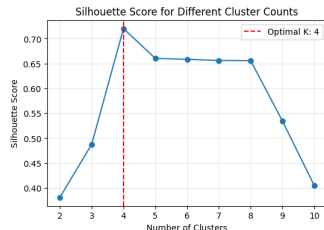
Principle: Measures how well-separated clusters are

For each sample i :

- ▶ $a(i)$ = average distance to other points in the same cluster
- ▶ $b(i)$ = average distance to points in the nearest different cluster
- ▶ Silhouette value $s(i) = \frac{b(i) - a(i)}{\max(a(i), b(i))}$
- ▶ **Selection criteria:** *Higher is better*

Interpretation:

- ▶ Range: $[-1, 1]$
- ▶ Close to 1: *Well-clustered*, Close to 0: *On cluster boundary*, Close to -1: *Not Well-clustered*.



Silhouette score visualization for different values of K

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Goal:

- ▶ Represent complex probability distributions as a mixture of Gaussians

Key advantages:

- ▶ Captures multimodal distributions that single Gaussians cannot model
- ▶ Provides probabilistic (soft) assignments to clusters
- ▶ Adapts to clusters of varying shapes, sizes, and densities

Applications:

- ▶ Market regimes in finance
- ▶ Clustering with soft assignments

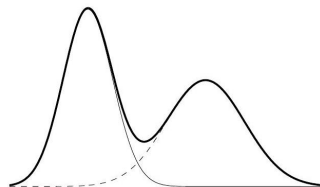
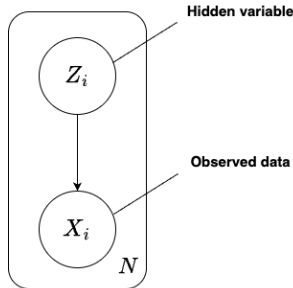


Figure: Density represented as a mixture of two Gaussians

Mathematical Formulation:

- ▶ n observations: $X_1, \dots, X_n \in \mathbb{R}^d$
- ▶ Hidden variables: $Z_1, \dots, Z_n \in \{1, \dots, K\}$
- ▶ For a mixture of K Gaussians:
 - ▶ $Z_i \sim \mathcal{M}(1, \pi_1, \dots, \pi_K)$
 - ▶ $(X_i | Z_i = k) \sim \mathcal{N}_d(\mu_k, \Sigma_k)$
 - ▶ Parameters $\theta = (\pi, \mu, \Sigma)$



Density function (via marginalization):

$$\begin{aligned} p_{\theta}(x_i) &= \sum_{z_i} p_{\theta}(x_i, z_i) = \sum_{z_i} p_{\theta}(x_i | z_i) p_{\theta}(z_i) = \sum_{k=1}^K p_{\theta}(x_i | z_i = k) p_{\theta}(z_i = k) \\ &= \sum_{k=1}^K \pi_k \mathcal{N}(x_i | \mu_k, \Sigma_k) \end{aligned}$$

Parameter Estimation:

- ▶ Need to learn $\theta = (\pi, \mu, \Sigma)$ using the **Expectation-Maximization algorithm**.
- ▶ Introduce a lower bound $\mathcal{L}(q, \theta)$ on the log-likelihood
- ▶ EM alternates between two steps:
 - ▶ Maximize wrt q (E-step)
 - ▶ Maximize wrt θ (M-step)
- ▶ Iterative process that guarantees improvement at each step

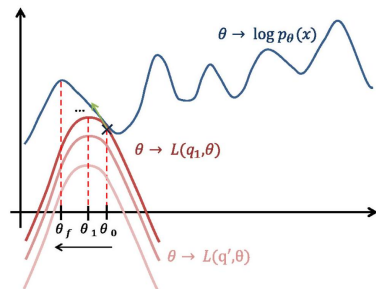


Figure: EM algorithm converging to a local maximum

Lower Bound via Jensen's Inequality:

For n observations with latent variables $z_1, \dots, z_n$:

$$\begin{aligned}\log p_{\theta}(x) &= \sum_{i=1}^n \log \left(\sum_{z_i} p_{\theta}(x_i, z_i) \right) \\ &= \sum_{i=1}^n \log \left(\sum_{z_i} q(z_i) \frac{p_{\theta}(x_i, z_i)}{q(z_i)} \right) \\ &\geq \sum_{i=1}^n \sum_{z_i} q(z_i) \log \left(\frac{p_{\theta}(x_i, z_i)}{q(z_i)} \right) = \mathcal{L}(q, \theta)\end{aligned}$$

Key Idea:

- ▶ **E-step:** Maximize $\mathcal{L}(q, \theta)$ with respect to q with fixed θ
- ▶ **M-step:** Maximize $\mathcal{L}(q, \theta)$ with respect to θ with fixed q

Algorithm EM Algorithm

Require: Data set $X = \{x_1, \dots, x_n\}$

Ensure: Optimal θ

- 1: **Initialization:** Choose initial parameters $\theta^{(0)}$.
- 2: **while** not converged **do**
- 3: **E-step:** Update q to maximize the lower bound wrt q .

$$q_{t+1} \in \arg \max_q (\mathcal{L}(q, \theta_t))$$

- 4: **M-step:** Update θ to maximize the lower bound wrt θ .

$$\theta_{t+1} \in \arg \max_{\theta} (\mathcal{L}(q_{t+1}, \theta))$$

- 5: **end while**
- 6: **return** Optimized parameters θ^* .

Finding optimal q :

- ▶ The gap between log-likelihood and lower bound is:

$$\begin{aligned} d &= \log p_{\theta}(x) - \mathcal{L}(q, \theta) \\ &= \sum_{i=1}^n D_{\text{KL}}(q(z_i) \parallel p_{\theta}(z_i | x_i)) \end{aligned}$$

- ▶ Gap is minimized when $q(z_i) = p_{\theta}(z_i | x_i)$ for all i
- ▶ The E-step is equivalent to setting q to be the posterior distribution:

$$q(z_i) = p_{\theta}(z_i | x_i)$$

- ▶ *Optional: Click here for the proof*

Key insight: We maximize the lower bound by setting q to match the posterior distributions under current parameter estimates

Rewriting the lower bound:

$$\mathcal{L}(q, \theta) = \sum_{i=1}^n \left(\sum_{z_i} q(z_i) \log p_{\theta}(x_i, z_i) - \sum_{z_i} q(z_i) \log q(z_i) \right)$$

For optimization with respect to θ :

- ▶ The second term $\sum_{z_i} q(z_i) \log q(z_i)$ doesn't depend on θ
- ▶ Maximizing $\mathcal{L}(q, \theta)$ with respect to θ is equivalent to:

$$\max_{\theta} \sum_{i=1}^n \sum_{z_i} q(z_i) \log p_{\theta}(x_i, z_i) = \max_{\theta} \mathbb{E}_{q(z)}[\log p_{\theta}(x, z)]$$

Key insight: The M-step maximizes the expected complete log-likelihood with respect to the posterior distribution

Algorithm EM algorithm

Require: Observations $x_1, \dots, x_n$

Ensure: Optimal θ

- 1: Initialize $\theta^{(0)}$
 - 2: **while** not converged **do**
 - 3: **E-step:** $q(z) = p(z|x; \theta^{(i-1)})$
 - 4: **M-step:** $\theta^{(i)} = \arg \max_{\theta} \mathbb{E}_q[\log p(x, z; \theta)]$
 - 5: $i \leftarrow i + 1$
 - 6: **end while**
-

Properties:

- ▶ Guaranteed to increase log-likelihood at each iteration
- ▶ Converges to a local maximum.

Problem: Use the EM algorithm to estimate parameters of a Gaussian Mixture Model.

- ▶ Given n observations $x_1, \dots, x_n \in \mathbb{R}^p$
- ▶ Assume latent variables $z_1, \dots, z_n$ with:
 - ▶ $z_i \sim \mathcal{M}(1, \pi_1, \dots, \pi_K)$ (multinomial with K outcomes)
 - ▶ $(x_i \mid z_i = j) \sim \mathcal{N}(\mu_j, \Sigma_j)$ (conditional Gaussian)
- ▶ Parameters to estimate: $\theta = (\pi, \mu, \Sigma)$ where:
 - ▶ $\pi = (\pi_1, \dots, \pi_K)$ are mixture weights
 - ▶ $\mu = (\mu_1, \dots, \mu_K)$ are component means
 - ▶ $\Sigma = (\Sigma_1, \dots, \Sigma_K)$ are covariance matrices

Task: Derive the E-step and M-step updates for this model.

Results of the exercise: *Click here for the detailed solution*

- **E-step:** Posterior probabilities $\tau_i^j = p_\theta(z_i = j | x_i)$

$$\tau_i^j = \frac{\pi_j \mathcal{N}(x_i | \mu_j, \Sigma_j)}{\sum_{j'=1}^K \pi_{j'} \mathcal{N}(x_i | \mu_{j'}, \Sigma_{j'})}$$

- **M-step:** Update parameters

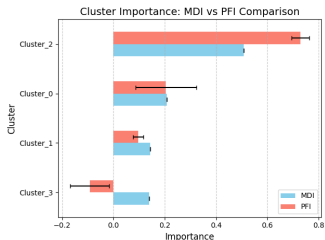
$$\begin{aligned}\pi_{j,t+1} &= \frac{1}{n} \sum_{i=1}^n \tau_i^j \\ \mu_{j,t+1} &= \frac{\sum_i \tau_i^j x_i}{\sum_i \tau_i^j} \\ \Sigma_{j,t+1} &= \frac{\sum_i \tau_i^j (x_i - \mu_{j,t+1})(x_i - \mu_{j,t+1})^T}{\sum_i \tau_i^j}\end{aligned}$$

Quiz Time!

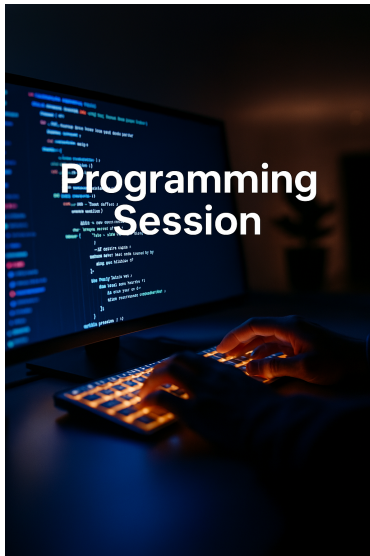
Click here to take the quiz

Feature Importance Analysis on a cluster level

- ▶ Generated dataset with 1000 samples
- ▶ 4 feature clusters with 4 features each:
 - ▶ One base feature per cluster (Base_i)
 - ▶ Three correlated features per cluster (Corr_{i-j})
- ▶ 5 pure noise features (Noise_i)
- ▶ Ground truth: Only "Base" features directly influence the target



- ▶ **Interpretation:**
- ▶ Cluster-level analysis correctly identifies noise clusters with negative importance.
- ▶ Feature importance at cluster level prevents underestimation of correlated features.



Programming Session 2: Section 3

- ▶ Section 3: Cluster-level Feature Importance Analysis
- ▶ *Click here to access the programming session*

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- ▶ *Click here to access the GitHub Page*

Feedback Poll

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Thank you for your attention